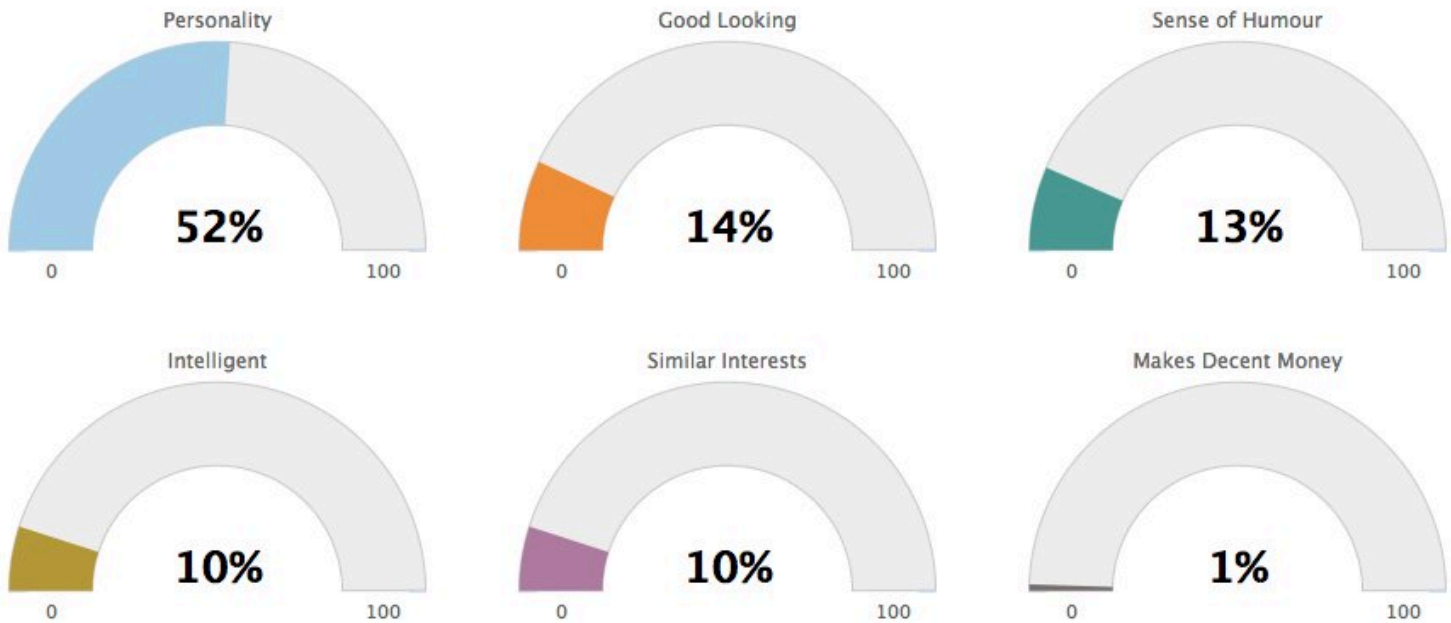


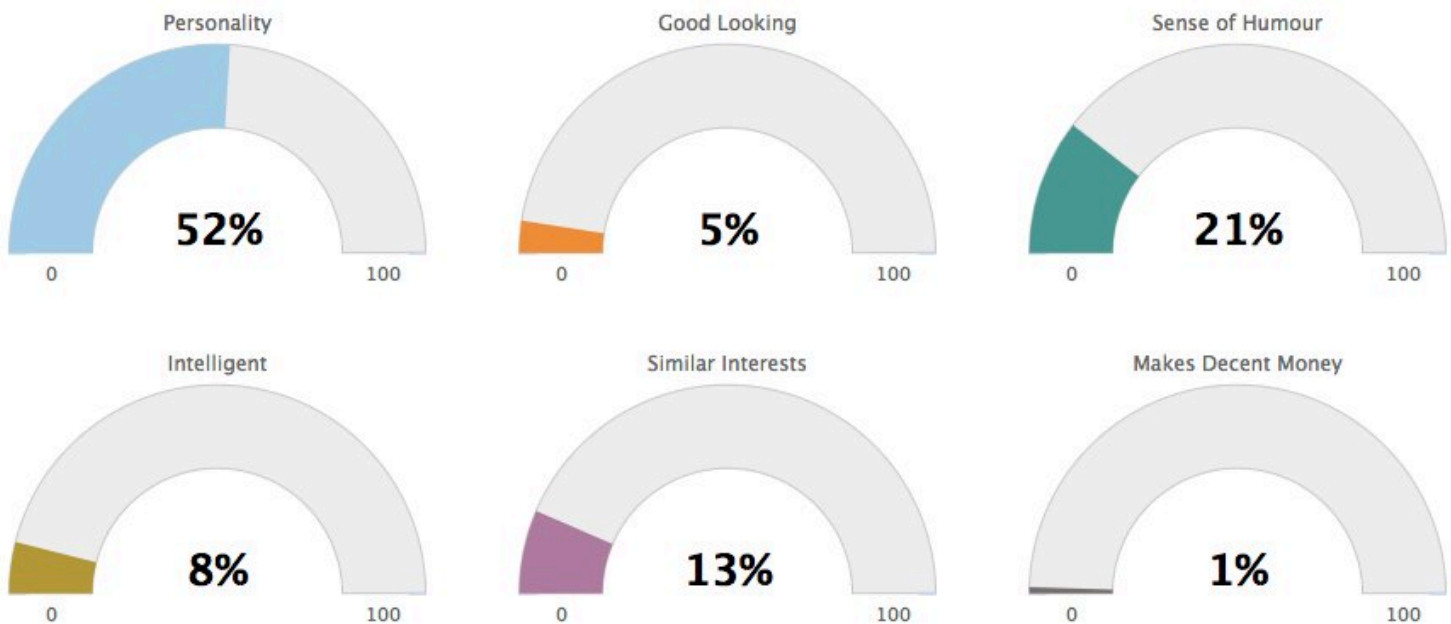
What characteristics are most important to British men and women?

Men and women both rate personality as the most important characteristic in a romantic partner. However, men rate good looks as the second most important characteristic while women prefer a good sense of humour.

Men



Women



Source: YouGov

In this case, I thought the gauge charts worked pretty well. Gauges are a default chart type in Highcharts, but Tableau doesn't do gauges out of the box. So, opportunist that I am, I figured that I'd write a quick blog post on

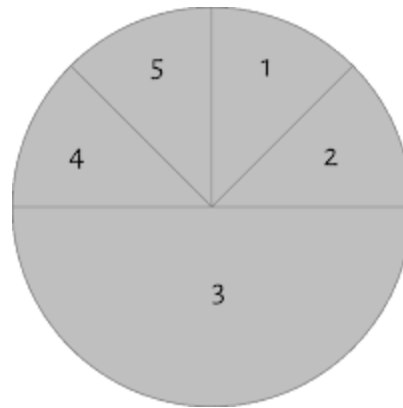
how to create charts like these in Tableau.

The Gauge

A few months ago, I wrote a blog on creating [NPS gauge charts in Tableau](#). When I saw Andy's visualization, it struck me that the Highcharts gauges were pretty similar to my NPS gauges. So, I figured I'd be able to pretty easily modify my NPS methodology to work for a percentage gauge. In case you haven't read my post on NPS gauges, let's start with observations about this chart.

First of all, it looks basically like a half of a donut chart (and donut charts in Tableau are essentially pie charts with a hole in the middle). The first slice of a pie chart in Tableau always starts at the 12:00 mark, but this chart would need to start at 9:00. The values of the chart will need to go from 0, starting at 9:00, to 100, ending at 3:00. This is the biggest challenge.

My approach creates a donut chart with 5 pre-defined slices, as shown below.



Slice 1 – Variable sized slice starting at 12:00 and ending between 12:00 and 3:00. This slice will be used to show the second part of the percentage (if the value is greater than 50). Otherwise, this slice will have 0 size, making it invisible.

Slice 2 – Variable sized slice starting and ending somewhere between 12:00 and 3:00, after slice 1. If the percentage is less than 100, then this slice will be visible and will appear in a light grey color.

Slice 3 – Hidden slice starting at 3:00 and ending at 9:00. This slice will always be the same color as the background, rendering it invisible, essentially making it a half-donut.

Slice 4 – Variable sized slice starting at 9:00 and ending between 9:00 and 12:00. This slice will be used to show the first part of the percentage (values between 0 and 50).

Slice 5 – Variable sized slice starting and ending somewhere between 9:00 and 12:00, after slice 4. If the percentage is less than 50, then this slice will be visible and will appear in a light grey color, like Slice 2.

The data set for this chart is exactly the same as the one I used for the NPS chart. It contains five records, one for each slice, with the following information about each:

Slice # – Numeric identifier of the slice (1, 2, etc.)

Slice – Name of the slice (Slice 1, Slice 2, etc.)

Details – Description of the slice and how it's used.

Strictly speaking, we could get by with just Slice # in the data set, but I decided to add a bit more information to make it less confusing to someone using the template.

There are no measures in the data set. We'll create the measure within Tableau, based on the percentage. Note: I'm assuming that you will have a percentage which can be fed to the visualization. For demonstration purposes, I'll just create a parameter allowing you to specify the percentage.

The Calculations

Calculating the pie chart size measure is a matter of some relatively simple math. To simplify my formulas, I created five calculated measures, one for each slice. Here are the formulas for each.

```
Value_Slice_1: IF [Percentage]>.5 THEN [Percentage]-.5 ELSE 0 END
Value_Slice_2: IF [Percentage]>.5 THEN 1-[Percentage] ELSE .5 END
Value_Slice_3: 1
Value_Slice_4: IF [Percentage]<.5 THEN [Percentage] ELSE .5 END
Value_Slice_5: IF [Percentage]<.5 THEN .5-[Value_Slice_4] ELSE 0 END
```

Note: "Percentage" is the name of the parameter noted earlier.

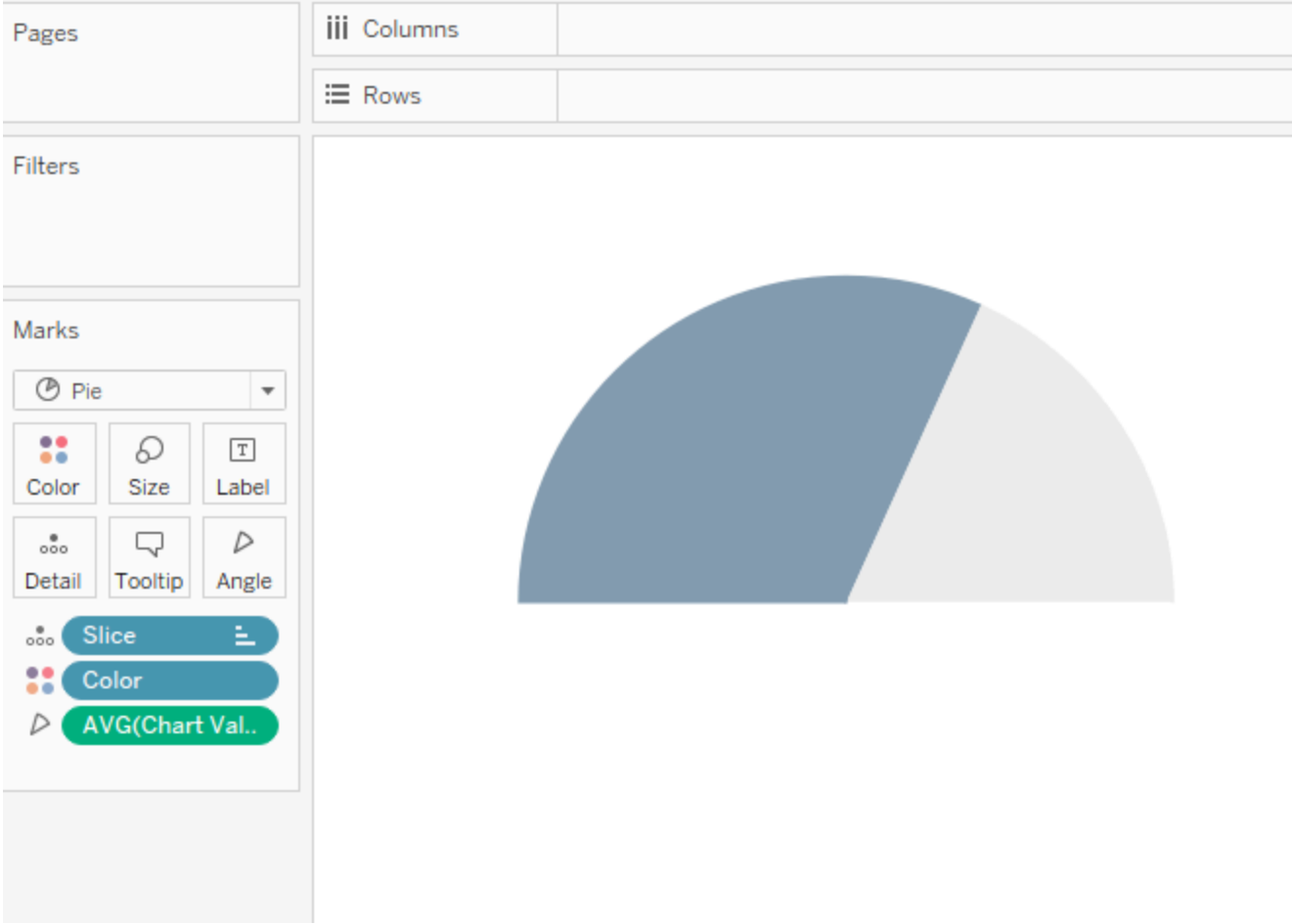
Then, I combine these values into a single measure, which I call **Chart Value**:

```
CASE [Slice #]
WHEN 1 THEN [Value_Slice_1]
WHEN 2 THEN [Value_Slice_2]
WHEN 3 THEN [Value_Slice_3]
WHEN 4 THEN [Value_Slice_4]
WHEN 5 THEN [Value_Slice_5]
END
```

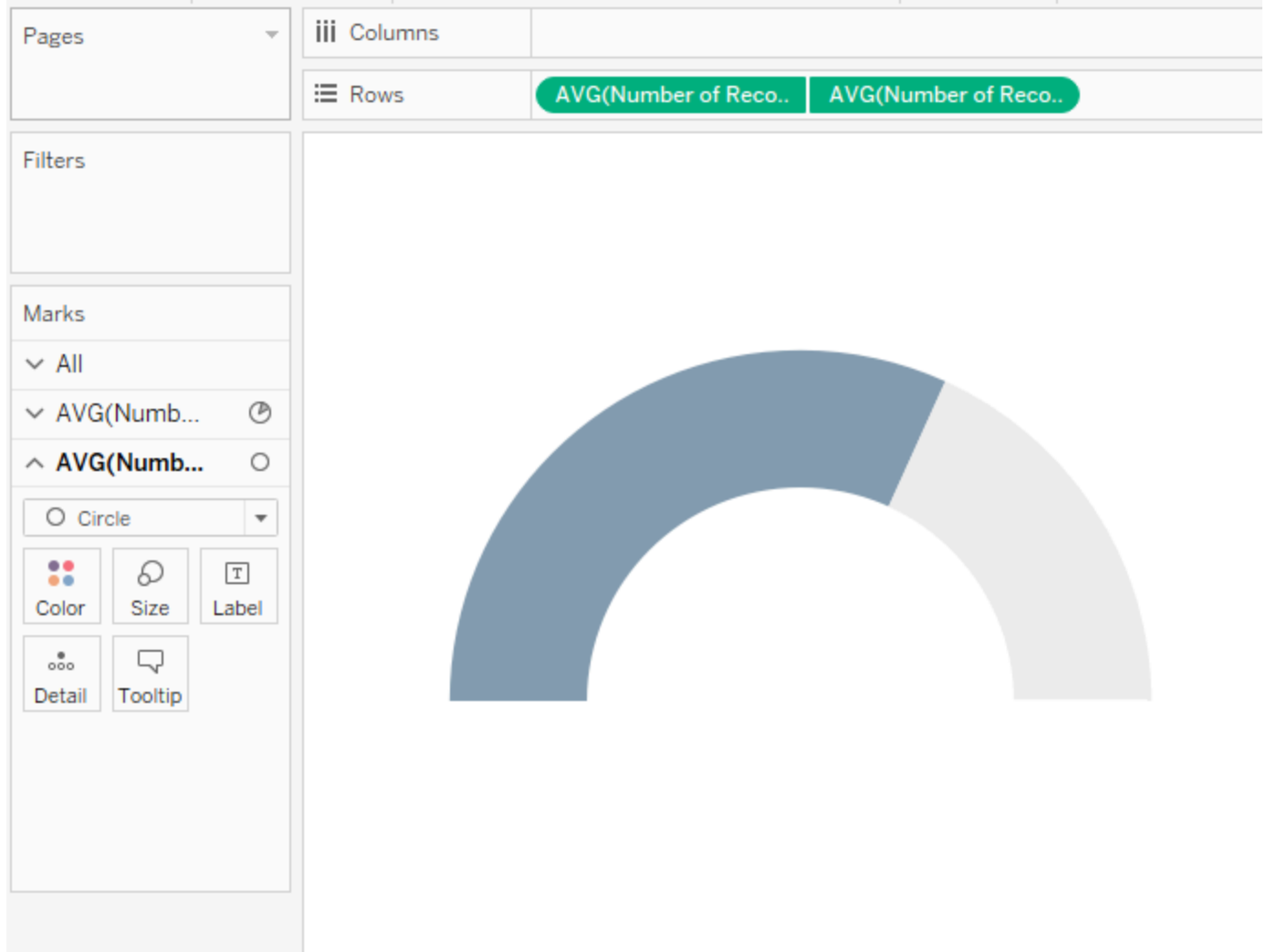
Finally, I create a calculated field called **Color**:

```
IF [Slice #]=3 then "Hidden"
ELSEIF [Slice #]=5 OR [Slice #]=2 THEN "Grey"
ELSE "Colored"
END
```

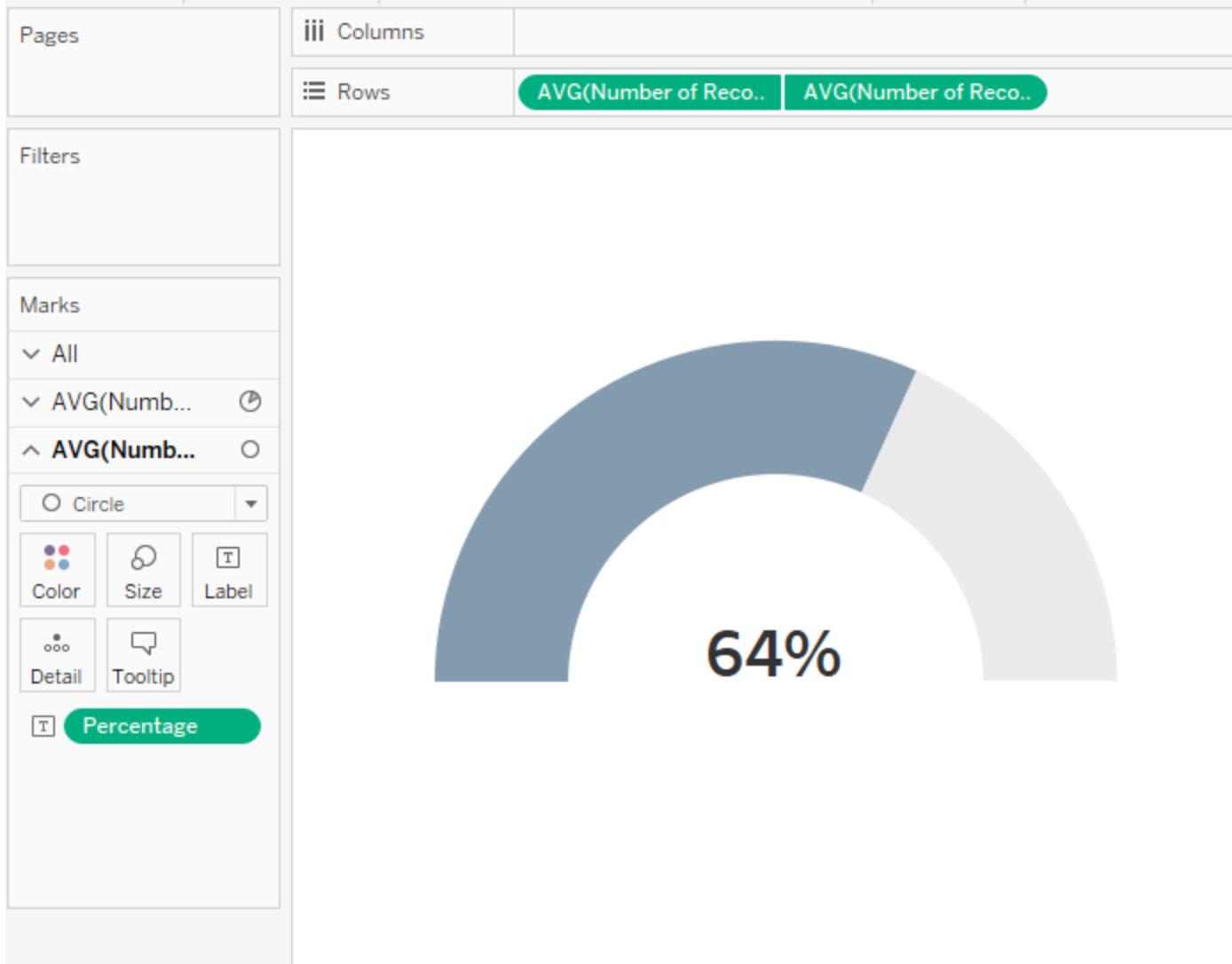
Next, based on these calculations, I create a very simple pie chart, as shown below:



I then change to a dual-axis chart, with the second axis showing a manually sized white circle.



Finally, I drag **Percentage** to the label card to add the text.



From here, you can add your sheet to a dashboard. In order prevent people from interacting with the individual slices of the pie, I'd recommend floating a blank over top of it.

Here's the final result:

Should You?...

Before I wrap this post, I think it's really important to address the question of whether or not you *should* use this type of chart. And, as always, the answer is "it depends." But, in most cases, you probably shouldn't. Gauges can be good when they are used to display a standalone KPI, as they are easy to understand. But they are pretty bad in situations where you're comparing multiple values. Our eyes simply aren't good at distinguishing the lengths of arcs, whereas our eyes are fantastic at distinguishing the lengths of 2 dimensional lines (i.e. bars). So, in these cases, you will definitely be better with a bullet chart or some other more linear visual. Plus, these are way easier to create in Tableau. For more on this topic, I'd recommend the following blogs:

[Alternatives to NPS Gauges](#) - My exploration of alternatives to gauges, including bullet charts.

[Accurate vs. Emotional Comparisons](#) - Zen Master, Steve Wexler, discusses the use of bars, pies, and other chart types and when to use them.

All this being said, I'd simply caution you if you're thinking about using gauges. I share this technique not because it's the way you should do it, but because there could be a time where it makes sense, for whatever reason. But you should avoid using a chart like this simply because it looks cool. While visual appeal is often important, you can still create visualizations with both visual appeal and strong analytical precision at the same time.

Required Files

If you have a need to create a gauge like this, then feel free to download my examples from Tableau Public. You can also find the Excel template [here](#).

Update March 15, 2019: A common problem with this blog has been an inability to make it work with your own data instead of the parameter. I apologize for that as I failed to properly explain how to do it. If you are struggling with this, I'd refer you to the following post on the Tableau Community Forum in which I show exactly how to do this: [Percentage Gauges with Your Own Data](#)

A New Technique (April 18, 2020)

The technique used previously required you to cross-join your data to my data set of five rows in order to create the five pie slices. Fortunately, I've since found that there is a much easier way to build this chart, which I'll share now. The same basic concepts exist so if you haven't read the full blog, that may be a worthwhile first step.

We'll start with a sample data file that has 10 customers with each of their percentages.

Customer #	Customer Name	Percentage
1100	Customer 1100	12%
1200	Customer 1200	0%
1300	Customer 1300	6%
1400	Customer 1400	29%
1500	Customer 1500	94%
1600	Customer 1600	100%
1700	Customer 1700	17%
1800	Customer 1800	26%
1900	Customer 1900	50%
2000	Customer 2000	33%

Unlike my previous method, this is the only data we need—we do not need the additional slice data set.

We'll connect to the data, then create the following calculated fields (feel free to just copy these from my workbook):

Slice 1

```
// Slice 1 size
// Slice starting at 0° and ending between 0° and 90°.
IF [Percentage]>0.5 THEN
    [Percentage]-0.5
ELSE
    0
END
```

Slice 2

```
// Slice 2 size
// Slice starting and ending somewhere between 0° and 90°, after slice 1.
IF [Percentage]>0.5 THEN
    1-[Percentage]
ELSE
    0.5
END
```

Slice 3

```
// Slice 3 size - Will always be 1
// Hidden slice starting at 90° and ending at 270°.
1
```

Slice 4

```
// Slice 4 size
// Slice starting at 270° and ending between 270° and 360°.
IF [Percentage]<0.5 THEN
    [Percentage]
ELSE
    0.5
END
```

Slice 5

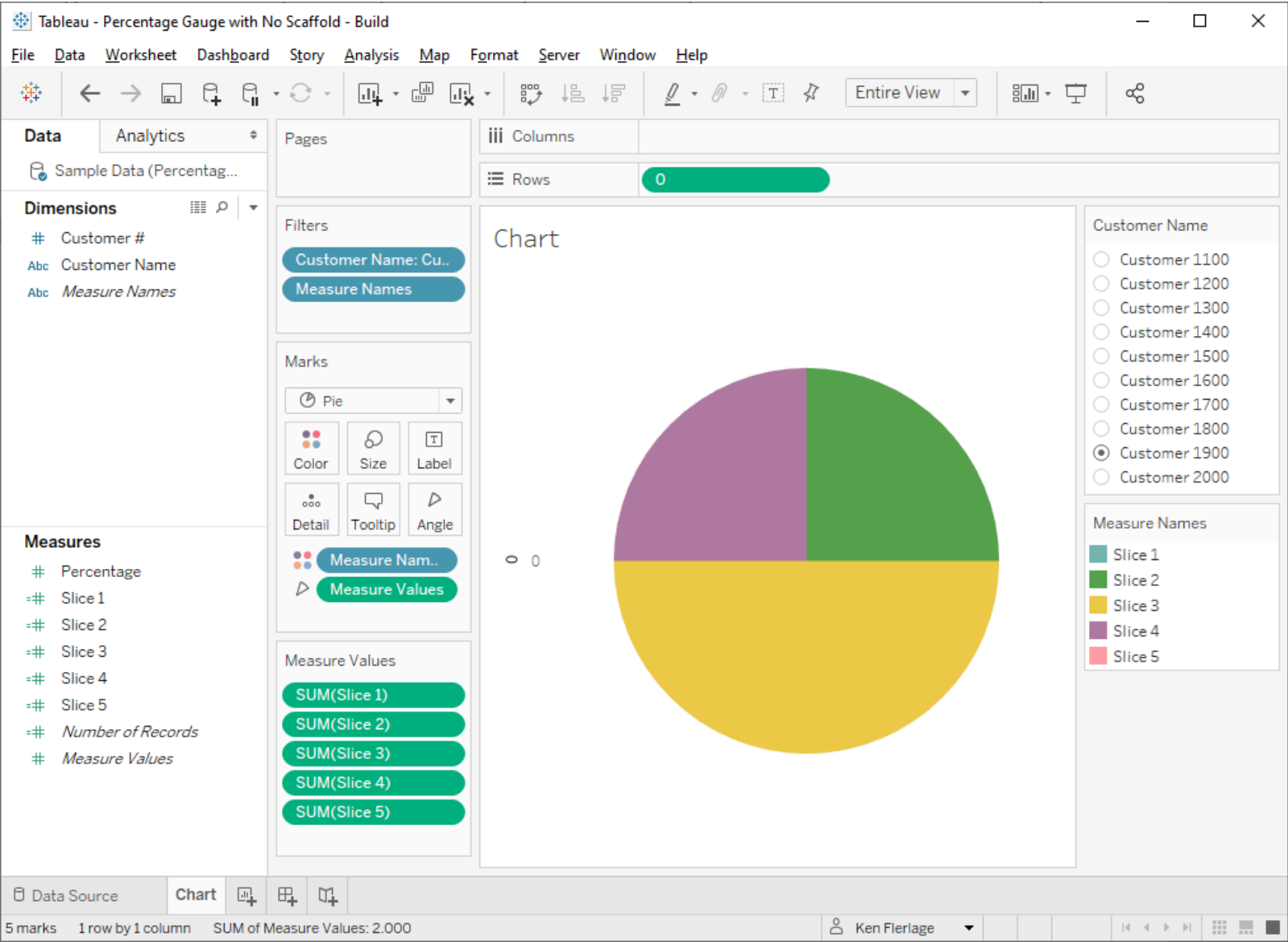
```
// Slice 4 size
// Slice starting and ending somewhere between 270° and 360°, after slice 4.
IF [Percentage]<0.5 THEN
```

```

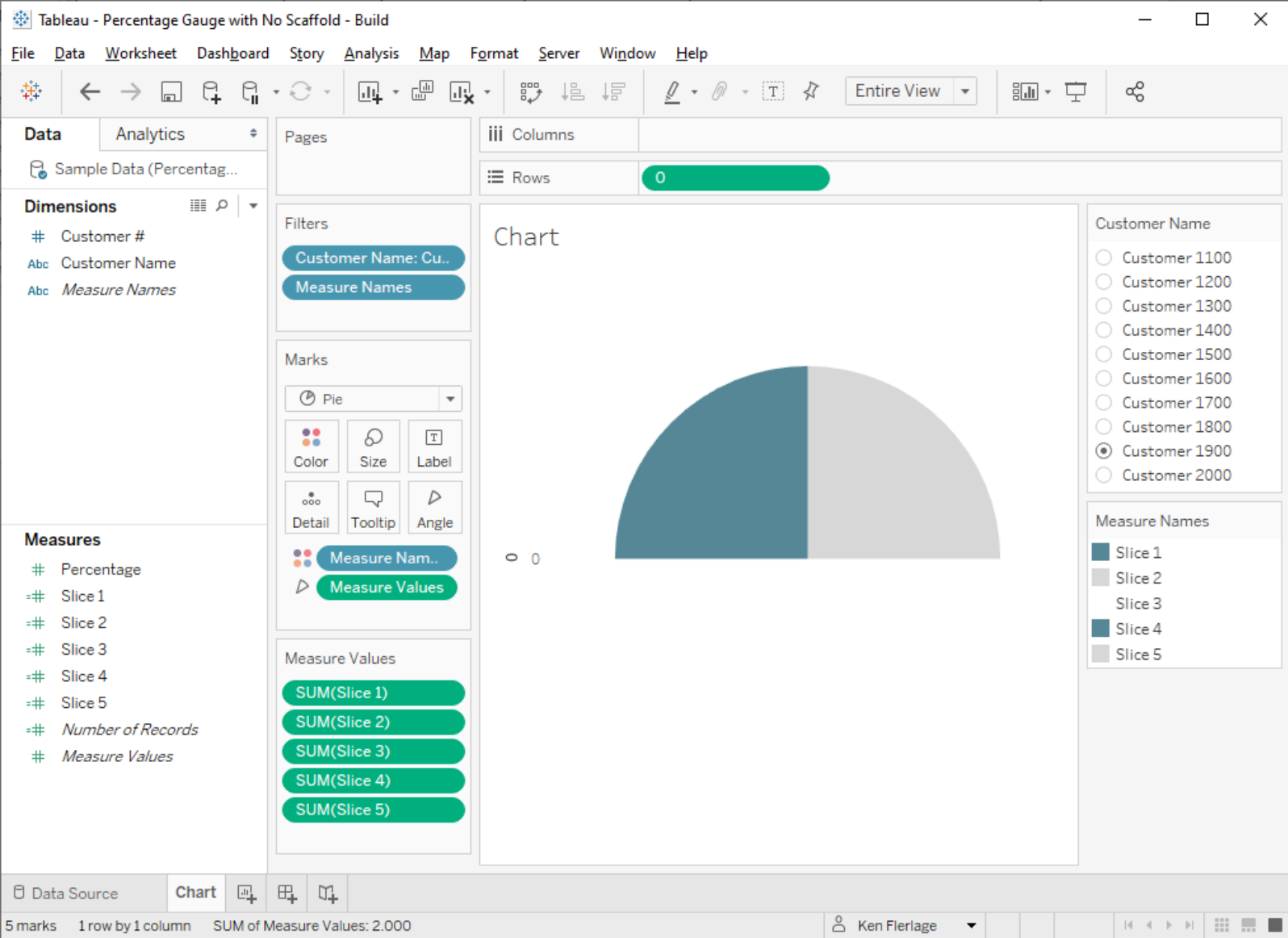
0.5-[Slice 4]
ELSE
0
END

```

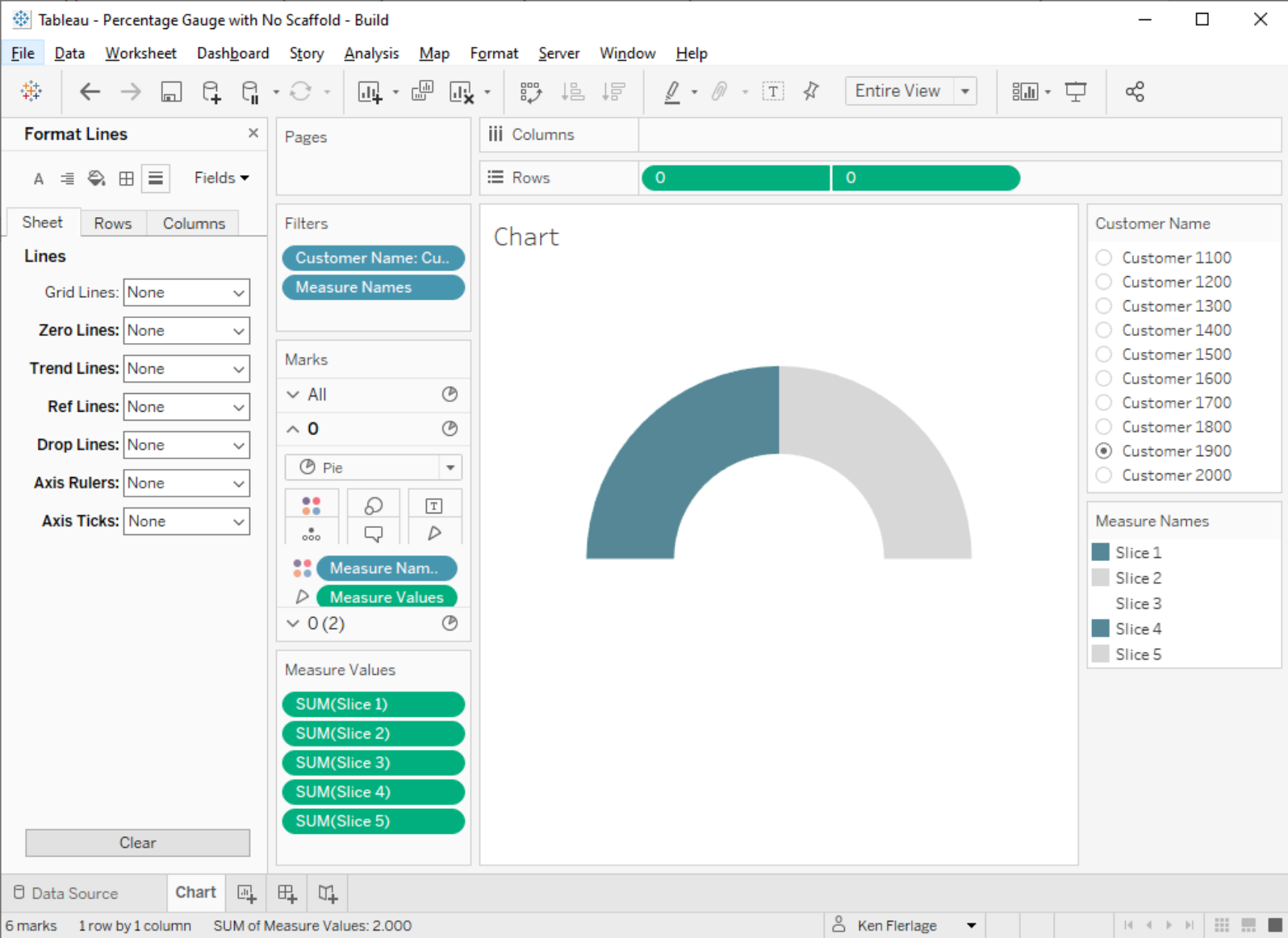
Now that we have all of our calculated fields, we'll create a "fake" axis on the rows shelf. Just type 0 into the shelf, then right-click the pill and change it to a dimension. Then drag **Measure Names** to the color card and **Measure Values** to the angle card. This will put *all* of your measures in the Measure Values pane—remove everything except the five slice measures.



Now we can set our colors. Set slice 3 to white (or whatever background color you're using), slices 2 and 5 to a shade of grey, and slices 1 and 4 to the color you wish your gauge to be.



Now duplicate the 0 pill on the rows shelf to create a new axis. Remove all pills from the marks card on this new axis. Change the mark type to “Circle”, adjust the size a bit so it is smaller than the pie chart, and change the color to white (or whatever background color you’re using). Then make it a dual axis and synchronize the axes.



Finally, drag **Percentage** to the label card of the second axis. Set the label to be aligned center/middle, then fiddle with the sizing of the text until you're satisfied.

